Newton's Laws with a Pulley Cart

U3 / S3.1

PREP — BEFORE DAY 1 AND DAY 2

WHEN	TASK	DETAIL
Days before	Charge and number	Charge 5 sensor carts and 5 GDX-FORs. Every group has one of each (confirmed by Matt, 2026-11-12). Number carts and sensors 1–5 so groups connect to their own pair.
Days before	Check signs	Roll one cart toward the pulley and pull its hook. Note whether velocity and force read positive. The handout takes toward the pulley as positive in Parts A and B, so F and a are positive and the Part A coasting slope is small and negative. If a cart reads negative toward the pulley, students change the sign as they record it. Also check that both the cart and the GDX-FOR show a pull on the hook as positive, as the Part C note on the handout says.
Days before	Test Part C yourself	**PROVISIONAL until Matt tests it.** (1) Where the string attaches on the back of the GDX-FOR (eyelet, tie point or strap slot). (2) Weigh a GDX-FOR on the balance. If its weight is close to the 0.5–1 N pull, it will droop between the cart hook and the string. A drooped sensor reads along a tilted axis, and the two readings drift apart. (3) Zeroing with the string slack zeroes the GDX-FOR in its drooped position, so the zero does not remove the tilt. See the backups below.
Before Day 1	Trays and tracks	5 tracks, a pulley clamped to one end, an end stop on the other. Tray per group: cart, mass bar, about 1.5 m of string, scissors, meter stick, hanger and slotted masses, folded towel. Balances within reach. **Run one Part A trial yourself** with 20 g and check the flat stretch is long enough to fit.
Before Day 2	GDX-FOR	One GDX-FOR per group, hook screwed in. Strings stay tied from Day 1 if the setups stay up.

Materials — total for 5 groups

Go Direct Sensor Cart × 5 · GDX-FOR × 5 (Day 2) · Dynamics Track × 5 · clamp-on pulley × 5 · end stop × 5 · cart mass bar × 5 · mass hanger and slotted masses, 20–100 g per group, × 5 · string, about 1.5 m × 5 plus spares · folded towel × 5 · meter stick and scissors × 5 · balance · student devices with Graphical Analysis × 5. **Matt to confirm:** the hanger's own mass. The handout uses 20, 40, 60 and 80 g with the hanger included. If the hanger alone is 50 g: Part A uses the hanger alone (50 g; the cart reaches about 1.3 m/s and coasts the 40 cm in about 0.3 s, still enough to fit), Part B uses 50, 60, 70 and 80 g (empty cart) and 50, 70 and 90 g (cart + bar). Track length is not confirmed either. The "start about 1 m from the pulley" step assumes a 1.2 m track or longer.

TIMING — TWO 50-MINUTE PERIODS (RECOMMENDED)

MIN	PHASE	WATCH FOR
D1 0–10	Pre-lab check, setup: pulley, level, string, loops	Strings cut to the wrong length. Pulley not adjusted to make the string level.
D1 10–20	Part A: 3 trials	Fitting across the speeding-up part. Pushing the cart on release.
D1 20–45	Part B: 4 runs empty, 3 runs with bar, F–a graph	Averages taken across the slack part. Bar added without re-massing.
D1 45–50	Reset	Masses back in the box. Setups can stay up for Day 2.
D2 0–5	Hook up the GDX-FOR, connect, zero both	Sensors zeroed with the hanger still pulling.
D2 5–20	Part C: 50 g and 100 g, held then moving	GDX-FOR drooping or twisted out of line.
D2 20–45	Post-lab in notebooks	Units on every number.

The handout's time flags add to about 65 minutes: setup 10, Part A 10, Part B 25, Part C 20. That is more than one period with the five-minute reset, which is why this is two days, as Matt approved. Day 2 has room for the whole post-lab in class.

FAILURE POINTS

- **String not level.** The string is kept level so the pull stays along the sensor axis and stays steady. Raise or lower the pulley (Setup step 6).
- **Hanging mass swinging.** A swinging hanger makes the force reading wobble. Let it hang still before releasing the cart.
- **Not zeroing with nothing pulling.** Zero with the hanger lifted, before every run. In Part C, zero both sensors that way. A leftover offset shifts the Part B intercept and the Part C comparison.
- **The string must go slack cleanly in Part A.** If the string is too long, the cart reaches the pulley before the mass lands and there is no flat part. If the mass bounces off a hard floor, the string snaps tight again and puts a spike in the flat part. That's what the towel and the Setup step 5 check are for.
- **Fit and average regions.** Part A fits only the flat part. Part B averages only the steady speeding-up stretch, leaving out the release spike and everything after the slack.
- **Catching late.** With 80 g the cart reaches about 1.5 m/s on 0.6 m of travel. Someone's hand stays near the pulley.
- **Part C sign.** Both sensors read positive, because each shows a pull on its own hook as positive. The two pulls point in opposite directions because of the third law: the GDX-FOR pulls the cart toward the pulley, and the cart pulls the GDX-FOR away from it. Students compare sizes, with no sign flipping.

Backups for Part C if the GDX-FOR droops

(a) **Hand pull, no pulley.** A student holds the GDX-FOR by hand and pulls the cart along the track by it. The two readings should still match, held or speeding up. Works for every group at once. (b) **Let it slide.** Let the GDX-FOR slide on the track with 100 g on the string, with the two hooks at the same height. **Matt to test** whether 100 g moves it smoothly. (c) **Ride it on a spare cart.** There are only 3 working spares (10 carts, 2 broken, 5 groups), so this can't cover every group at once. (d) **GDX-FOR in the vertical string**, between the string and the hanger. The cart sensor reads more than the GDX-FOR, by about the GDX-FOR's weight. The string passes the pull along, but this is not a 3rd-law pair, so the Part C wording on the handout would need rewriting.

STRING-LENGTH ESTIMATE — CHECK

The string is a horizontal piece (hook to pulley) plus a hanging piece (pulley to hanger). The hanger lands when the hanging piece is about the pulley height H. Whatever is left is the horizontal piece, which is how far the cart still is from the pulley. So for 40 cm of coasting room, $L \approx H + 0.40$ m, wherever the cart starts. Example: H = 0.95 m gives L = 1.35 m. Starting 1.00 m out, the hanger starts 0.35 m below the pulley and falls 0.60 m. The cart speeds up for 0.60 m and then has 0.40 m to coast. The hanger's own height only adds coasting room, so the check in Setup step 5 says "at least 40 cm." **A tempting version, $L \approx d + H - 0.40$ m, is wrong:** it makes the mass fall only 40 cm and leaves the cart d − 0.40 m from the pulley.

EXPECTED RESULTS — SAMPLE DATA

Sample data, not real measurements. These come from a model of the setup with a cart of about 0.28 kg and rolling friction of about 0.03 N, with realistic scatter added. The handout's example rows are rows from this set. Replace the masses with your own balance readings.

QUANTITY	ACCEPTED	SAMPLE	TYPICAL RANGE
Part A slope of flat part	0 with no friction	−0.094, −0.108, −0.097 m/s²	about −0.05 to −0.15 m/s²
Part B slope, cart only	0.282 kg (sample balance)	0.292 kg · 3.5%	within about 10%
Part B slope, cart + bar	0.779 kg (sample balance)	0.744 kg · 4.5%	within about 10%
Part C percent difference	0%	1.0% to 3.8%	under about 5% when lined up

WORKED SAMPLE ANALYSIS — SAMPLE DATA, NOT REAL MEASUREMENTS

Sample cart 0.282 kg, sample bar 0.497 kg, so cart + bar = 0.779 kg. g = 9.8 m/s².

Part A — first law (20 g)

TRIAL	HANGING MASS (g)	V AT START OF FLAT (m/s)	V AT END OF FLAT (m/s)	SLOPE (m/s²)
1	20	0.81	0.78	−0.094
2	20	0.83	0.79	−0.108
3	20	0.80	0.77	−0.097

mean slope = $\frac{-0.094 + (-0.108) + (-0.097)}{3} = \frac{-0.299\ \text{m/s}^2}{3} = -0.0997\ \text{m/s}^2 \approx -0.100\ \text{m/s}^2$

For you, not asked on the handout: friction f = m|a| = (0.282 kg)(0.0997 m/s²) = 0.028 N. That is close to the Part B cart-only intercept, 0.023 N.

Part B — second law

RUN	CART SETUP	HANGING MASS (g)	F (N)	a (m/s²)
1	cart only	20	0.18	0.53
2	cart only	40	0.33	1.06
3	cart only	60	0.48	1.57
4	cart only	80	0.60	1.97
5	cart + bar	40	0.37	0.39
6	cart + bar	60	0.53	0.61
7	cart + bar	80	0.69	0.82

Least-squares fits: cart only, slope 0.292 kg, intercept +0.023 N. Cart + bar, slope 0.744 kg, intercept +0.079 N.

Cart only: percent error = $\frac{|0.292\ \text{kg} - 0.282\ \text{kg}|}{0.282\ \text{kg}} \times 100 = \frac{0.010\ \text{kg}}{0.282\ \text{kg}} \times 100 =$ **3.5%**

Cart + bar: percent error = $\frac{|0.744\ \text{kg} - 0.779\ \text{kg}|}{0.779\ \text{kg}} \times 100 = \frac{0.035\ \text{kg}}{0.779\ \text{kg}} \times 100 =$ **4.5%**

Slope rise: 0.744 kg − 0.292 kg = 0.452 kg, against 0.497 kg added (9% low). The intercept is bigger with the bar on because a heavier cart has more rolling friction.

Part C — third law · PROVISIONAL until Matt tests the setup

RUN	HANGING MASS (g)	HELD OR MOVING	CART SENSOR (N)	GDX-FOR (N)	PERCENT DIFFERENCE
1	50	held	0.49	0.48	2.1%
2	50	moving	0.27	0.26	3.8%
3	100	held	0.98	0.97	1.0%
4	100	moving	0.49	0.50	2.0%

Run 1: average = $\frac{0.49\ \text{N} + 0.48\ \text{N}}{2} = 0.485\ \text{N}$ percent difference = $\frac{|0.49\ \text{N} - 0.48\ \text{N}|}{0.485\ \text{N}} \times 100 = \frac{0.01\ \text{N}}{0.485\ \text{N}} \times 100 =$ **2.1%**

Run 2: $\frac{0.01\ \text{N}}{0.265\ \text{N}} \times 100 =$ **3.8%** Run 3: $\frac{0.01\ \text{N}}{0.975\ \text{N}} \times 100 =$ **1.0%** Run 4: $\frac{0.01\ \text{N}}{0.495\ \text{N}} \times 100 =$ **2.0%**

Held check: W = mg = (0.050 kg)(9.8 m/s²) = 0.49 N and (0.100 kg)(9.8 m/s²) = 0.98 N, matching the held readings.

Moving rows are modelled with an **assumed** GDX-FOR mass of 0.20 kg, held level with no friction (not measured; weigh yours). a = (mg – f) ÷ (cart + GDX-FOR + hanging mass) gives 0.87 m/s² for 50 g and 1.64 m/s² for 100 g. The pull between the sensors only accelerates the cart and beats its friction: (0.282 kg)(0.87 m/s²) + 0.027 N = 0.27 N. While moving, the string tension is less than mg and more than either sensor reading: T = m(g – a) = 0.45 N for 50 g (mg = 0.49 N) and 0.82 N for 100 g (mg = 0.98 N).

PRE-LAB KEY

1. Newton's three laws. (1) An object stays at rest or keeps moving at a constant velocity unless a net force acts on it. (2) ∑F = ma: net force causes acceleration; more force, more acceleration; more mass, less acceleration. (3) When A exerts a force on B, B exerts a force of the same size in the opposite direction on A. Full credit for each law in the student's own words with its key idea.
2. String length with H = 0.95 m. 0.95 m + 0.40 m = 1.35 m, loop to loop.
3. Predicted v–t graph. A straight line rising from zero (speeding up at a steady rate), then a nearly flat line starting where the string goes slack, sloping down very slightly, then a sharp drop at the catch. Credit needs the rise, the flat part and the slack label.

POST-LAB KEY — FROM THE SAMPLE DATA

1. First law, Part A. Once the mass landed, the string went slack and nothing pulled the cart forward, so the net force was about zero. Also accept "small and backward (friction), about 0.03 N." The velocity stayed nearly constant (0.81 to 0.78 m/s), as the first law says. It was not perfectly flat: the slope was about –0.10 m/s² because rolling friction still acted backward. No credit for "the force from the string kept it going." That is the misconception this part exists to fix.
2. Second law, Part B. As F went up, a went up: the points rose in a straight line. With the bar on, the same F gave a smaller a, so the line was steeper. Its slope, the mass, went from 0.292 to 0.744 kg.
3. Percent error. 3.5% (cart only) and 4.5% (cart + bar). Work is above.
4. Error source and direction. Needs a specific cause and the direction it pushed the mass. Good answers: the averaged stretch ran into the release spike or the slack part, pulling points off the line; friction that grows with speed, so faster runs need extra force and the slope comes out high; the wheels' rotational inertia, which adds to the effective mass, so the slope comes out high; sensor calibration, which scales every F reading. Note: "forgot to zero" or "track not level" shifts the intercept, not the slope, so it does not explain a wrong mass. Partial credit only. "Human error" earns nothing.
5. Third law, Part C. The two readings were the same size at every moment (within about 0.01 N), held or moving. The GDX-FOR pulled the cart toward the pulley, and the cart pulled the GDX-FOR back toward the cart: equal size, opposite direction, acting on different objects.
6. Percent difference. 2.1%, 3.8%, 1.0%, 2.0%; work above.

TEMPLATE EXCEPTIONS

This lab departs from the locked lab template: no Safety section, no Disposal/Alconox block, filled example tables, and a reduced pre-lab and post-lab. These exceptions are approved and recorded in SHULL-CHG-0073.

GRADING

No point split for Physics labs is set yet. Matt sets this. The carbonless copy is graded individually.

MISSED LAB

A student who misses a day completes a Missed Lab Analysis rather than re-running the lab. Give them one group's Tables 1–3 from the same period and a screenshot of that group's Part A v–t graph and F–a graph. They answer all six post-lab questions from that data.